

LNPTM THERMOCOMPTM COMPOUND RC004SXS

RC-1004 HS

REGION AMERICAS

DESCRIPTION

LNP THERMOCOMP RC004SXS is a compound based on Nylon 66 containing 20% Carbon Fiber. Added feature of this grade includes: Heat Stabilized, Electrically Conductive.

TYPICAL PROPERTY VALUES

Revision 20200311

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, brk, Type I, 5 mm/min	222	MPa	ASTM D 638
Tensile Strain, brk, Type I, 5 mm/min	2.2	%	ASTM D 638
Tensile Modulus, 50 mm/min	16680	MPa	ASTM D 638
Flexural Stress, brk, 1.3 mm/min, 50 mm span	348	MPa	ASTM D 790
Flexural Modulus, 1.3 mm/min, 50 mm span	13700	MPa	ASTM D 790
Tensile Stress, break, 5 mm/min	242	MPa	ISO 527
Tensile Strain, break, 5 mm/min	2.5	%	ISO 527
Tensile Modulus, 1 mm/min	16740	MPa	ISO 527
Flexural Stress	331	MPa	ISO 178
Flexural Modulus, 2 mm/min	12920	MPa	ISO 178
IMPACT			
Izod Impact, unnotched, 23°C	884	J/m	ASTM D 4812
Izod Impact, notched, 23°C	67	J/m	ASTM D 256
Multiaxial Impact	2	J	ISO 6603
Instrumented Impact Total Energy, 23°C	6	J	ASTM D 3763
Izod Impact, unnotched 80*10*4 +23°C	49	kJ/m ²	ISO 180/1U
Izod Impact, notched 80*10*4 +23°C	6	kJ/m ²	ISO 180/1A
THERMAL			
HDT, 0.45 MPa, 3.2 mm, unannealed	260	°C	ASTM D 648
HDT, 1.82 MPa, 3.2mm, unannealed	252	°C	ASTM D 648
CTE, -30°C to 30°C, flow	2.1E-05	1/°C	ASTM D 696
CTE, -30°C to 30°C, xflow	9.0E-05	1/°C	ASTM D 696
HDT/Bf, 0.45 MPa Flatw 80*10*4 sp=64mm	260	°C	ISO 75/Bf
HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm	249	°C	ISO 75/Af
PHYSICAL			
Specific Gravity	1.23	-	ASTM D 792
Moisture Absorption, 50% RH, 24 hrs	0.89	%	ASTM D 570
Mold Shrinkage, flow, 24 hrs	0.2 – 0.4	%	ASTM D 955
Mold Shrinkage, xflow, 24 hrs	0.9 – 2	%	ASTM D 955
Density	1.23	g/cm ³	ISO 1183
Moisture Absorption (23°C / 50% RH)	1.2	%	ISO 62
INJECTION MOLDING			
Drying Temperature	80	°C	

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Drying Time	4	hrs	
Maximum Moisture Content	0.15 – 0.25	%	
Melt Temperature	280 – 305	°C	
Front - Zone 3 Temperature	295 – 305	°C	
Middle - Zone 2 Temperature	280 – 295	°C	
Rear - Zone 1 Temperature	265 – 275	°C	
Mold Temperature	95 – 110	°C	
Back Pressure	0.2 – 0.3	MPa	
Screw Speed	30 – 60	rpm	

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